



GE HEALTHCARE

Technical Publications

GE Part Number 5890114
Revision 1

TEAL PDU Module in XGD3 Cabinet

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Operating Documentation

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IMPORTANT SAFETY INSTRUCTIONS

Even when an EPO/EMO switch is pressed this does not guarantee that the power has been cut off coming into the unit.

To avoid tipping over, do not tip beyond 10° from vertical and avoid abrupt movements.



⚠ DANGER



Voltages in excess of 480VAC are present within this equipment. Turn power off at wall feeder breaker before servicing. Failure to do so could result in serious injury or death.

Des tensions supérieures à 480 V sont présentes dans cet équipement. Coupez l'alimentation au disjoncteur d'alimentation murales avant de réparer. Ne pas le faire pourrait entraîner des blessures graves ou la mort.



⚠ DANGER



Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.

Risque de choc électrique. Verrouiller alimentation électrique au panneau principal. Ne pas le faire pourrait entraîner des blessures graves ou la mort

⚠ WARNING

Failure to set transformer input taps to match nominal input voltage will result in improper output voltages which may damage equipment, cause load equipment to malfunction or cause injury.

Si vous ne réglez les robinets d'entrée du transformateur pour correspondre à la tension d'entrée nominale se traduira par des tensions de sortie irrégulières qui peuvent endommager l'équipement, l'équipement en charge provoquer un dysfonctionnement ou provoquer des blessures.

⚠ WARNING

Failure to set control transformer to match nominal input voltage may result in equipment damage or malfunction of equipment.

Si vous ne réglez transformateur de commande pour correspondre à la tension d'entrée nominale peut entraîner des dommages matériels ou un dysfonctionnement de l'équipement.

⚠ IMPORTANT

Power conditioner must be properly grounded for safety per National Electric Code, State or Local code.

Conditionneur de puissance doit être correctement mise à la terre pour la sécurité par le National Electric Code, État ou locale de code.

⚠ WARNING

No modification of this equipment is allowed.

Aucune modification de cet équipement est autorisé.

⚠ WARNING

To avoid risk of electric shock, this equipment must only be connected to supply mains with protective earth.

Pour éviter tout risque de choc électrique, cet équipement ne doit être raccordée au réseau électrique avec la terre de protection.

⚠ WARNING

Connecting electrical equipment to MSOs effectively leads to creating an ME system, and can result in a reduced level of safety.

Connexion d'un équipement électrique pour les OSM conduit effectivement à la création d'un système de ME, et peut résulter en une diminution du niveau de sécurité.

⚠ WARNING

All operators of this unit must have "High Voltage" service training.

Tous les opérateurs de cet appareil doivent avoir une formation "High Voltage" de service.

⚠ WARNING

Unit is to be used solely for the purpose in which it was designed.

Unité doit être utilisée uniquement dans le but dans lequel il a été conçu.

Note: To safely terminate operation of the ME equipment, turn off the load, then turn the main breaker off.

Pour en toute sécurité mettre fin au fonctionnement de l'équipement ME, éteignez la charge, puis éteindre le disjoncteur principal.

Note: The supply mains isolation is provided by the CB1 circuit breaker on all poles simultaneously.

L'isolement d'alimentation secteur est fourni par le disjoncteur CB1 simultanément sur tous les pôles. Cette note conforme à la norme AAMI ES60601-1 partie 1 3ème édition.

Note: Assembly of ME systems and modifications during actual service life shall be evaluated based on the requirements of standard AAMI ES60601-1 part 1 3rd edition.

Assemblée des systèmes ME et modifications au cours de la durée de vie réelle doit être évaluée sur la base des exigences de la norme AAMI ES60601-1 partie 1 3ème édition.

Note: The equipment must be installed such that enough space is provided in front of the unit to allow for proper access to perform maintenance.

L'équipement doit être installé de telle sorte qu'un espace suffisant soit fourni devant l'unité pour permettre un accès approprié pour effectuer la maintenance.

Note: Follow proper local and national disposal requirements and codes when the life cycle ends for the equipment.

Suivez les exigences et les codes d'élimination locaux et nationaux appropriés lorsque le cycle de vie de l'équipement se termine.

Warning Labels:



Line voltage present with machine power off. Risk of electric shock or burn. Turn off & lock out facility power before servicing.

Tension présente ligne avec alimentation de la machine hors tension. Risque de choc électrique ou de brûlure. Éteignez et verrouillez l'alimentation avant l'entretien installation.



Hazardous voltage. Disconnect power before servicing.

Tension dangereuse. Débranchez l'alimentation avant l'entretien.



To reduce the risk of electric shock, do not remove cover. Refer servicing to qualified personnel.

Pour réduire le risque de choc électrique, ne pas enlever le couvercle. Confiez l'entretien à un personnel qualifié.



Class1 ordinary equipment is not suitable for use in the presence of a flammable anesthetic mixture with air or oxygen or nitrous oxide. Rated for continuous operation.

Class1 équipement ordinaire n'est pas adapté pour une utilisation en présence d'un mélange anesthésique inflammable avec l'air ou d'oxygène ou d'oxyde nitreux. Prévu pour fonctionner en continu.



Follow operating instructions. On ME Equipment Follow instructions for use.

Suivez les instructions de fonctionnement. Le ME Equipement Suivez les instructions pour l'utilisation.

IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1.1 General Description

The XGD3 Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MR system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among 380/400/415/480 VAC at a rated frequency of 50/60 Hz. The power rating for this PDU is 79 kVA continuous and 103 kVA instantaneous. The PDU mounts into the GE 5720999-200 cabinet.

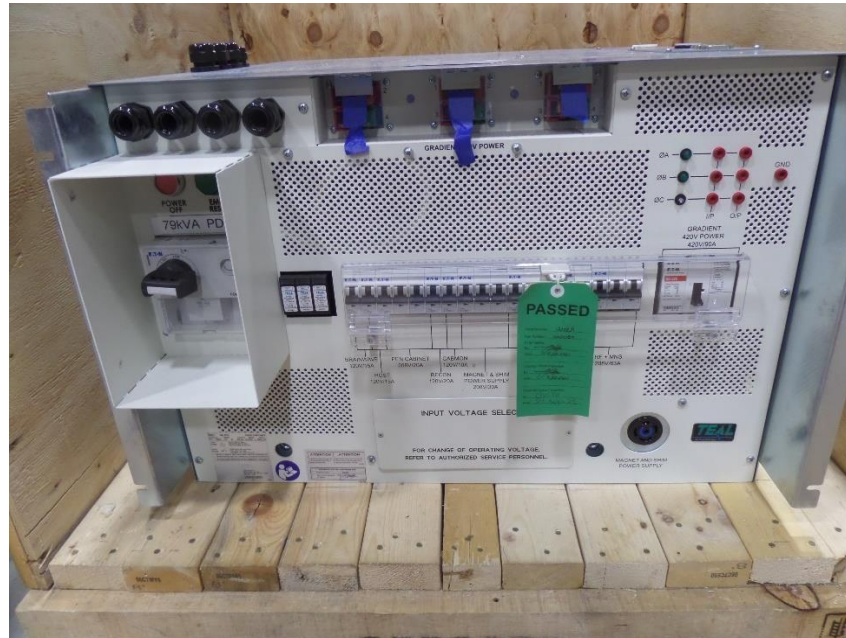
The design of the MR System is such that there will be no interaction with the PDU during operation by anyone. Not the operator, not the patient and not by anyone else.

This unit can only be operated from the front of the equipment rack that the unit is installed in.

Electromagnetic interference does not affect the operation of this unit.

For questions on the specifics of this unit contact:

MTE Corporation
N83 W12650 Stratton Circle, Suite 100
Menomonee Falls, WI 53051



**XGD3 Power Distribution Unit
Illustration 1-1**

1.2 Scope

This manual describes procedures for installation and basic operation of the XGD3 Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2- SPECIFICATIONS

GE Specifications: 5890114PSP

GE Part Number
5890114

2.1 Electrical Specifications

Power Rating

79 kVA Continuous; 103 kVA Instantaneous

Input Voltage

380/400/415/480 VAC Delta.

Output Voltage

208/120 VAC WYE and 242/420 VAC WYE Output

Circuit Breaker and Connection Requirements

Breaker Size	Function	Ref. Desig.	Sec Voltage	Wire to Phase	E-Stop	Customer Access	Connection Type
90A 3 Pole	Gradient 420V Power	CB2	420	ABC	Yes	Front	APP#1321G4, G5 (3 Connectors)
63A 3 Pole	RF + MNS	CB3	208	ABC	Yes	Right Side	Breaker
20A 3 Pole	AC Dist	CB4	208	ABC	Yes	Top Left Side	Breaker
15A 1 Pole	Brainwave	CB5	120	A	No	Top Left Side	Breaker
15A 1 Pole	Host	CB6	120	A	No	Top Left Side	Breaker
40A 3 Pole	System Cooling	CB7	208	ABC	No	Right Side	Breaker
30A 3 Pole	Magnet and Shim Power Supply	CB8	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle, front of cab)
10A 2 Pole	Gradient 48V Control	CB9	208	BC	No	Top	NEMA 6-20R
20A 1 Pole	Recon	CB10	120	B	No	Top	NEMA 6-20R
10A 1 Pole	CabMon	CB11	120	B	No	Top	NEMA 6-20R

Overload Protection

Input main circuit breaker (CB1). Output distribution circuit breakers.

Load Current

84 Amps total maximum continuous (120/208V)

68 Amps total maximum continuous (420V), 134A Peak 5 sec Max.

Transformer Impedance

Less than 2%

Efficiency

98% Typical

2.2 Environmental Specifications**Heat Dissipation**

4,076 BTU/Hr

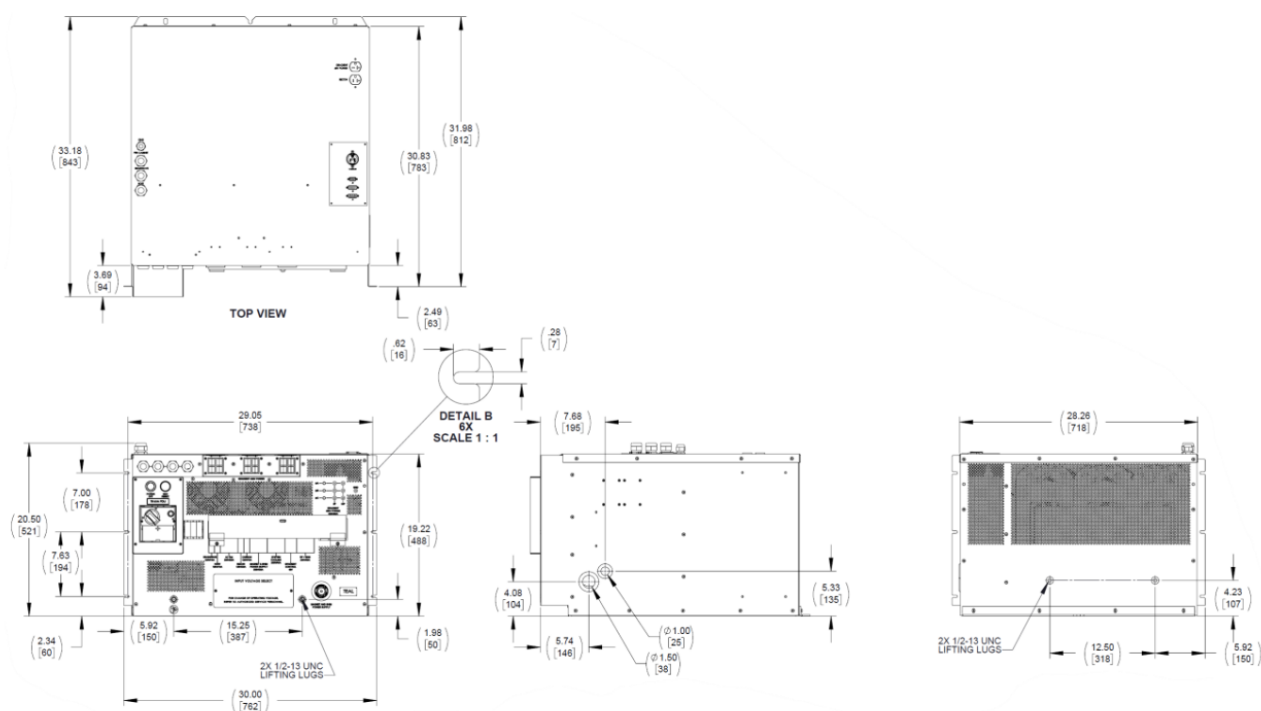
2.3 Mechanical Specifications**Weight**

879 lbs. / 398 kg.

Other environmental Specs

<i>Operating Humidity Range</i>	20-75% non-condensing
<i>Operating Atmospheric Pressure Range</i>	70-106kPa
<i>Transport and storage Temp Range</i>	-20°C to 50°C
<i>Transport and Storage Humidity Range</i>	10-90% non-condensing
<i>Transport and Storage Atmospheric Pressure Range</i>	70-106kPa
<i>Max Temperature Gradient</i>	≤3°C / hr
<i>Thermal Cutout (N/C) Temperature</i>	150°C

Dimensions



XGD3 Overall Dimensions Illustration 2-1

3- INSTALLATION AND OPERATION

3.1 Installation

The installation location must meet the following criteria:

Ambient temp between 0°F and 104°F

Humidity between 0% and 90% non-condensing

Altitude below 6000 feet

Pollution degree 2

Overvoltage Category II

Installation must be performed by a licensed electrician or a factory qualified field engineer.

Unit is intended for use in a controlled environment.



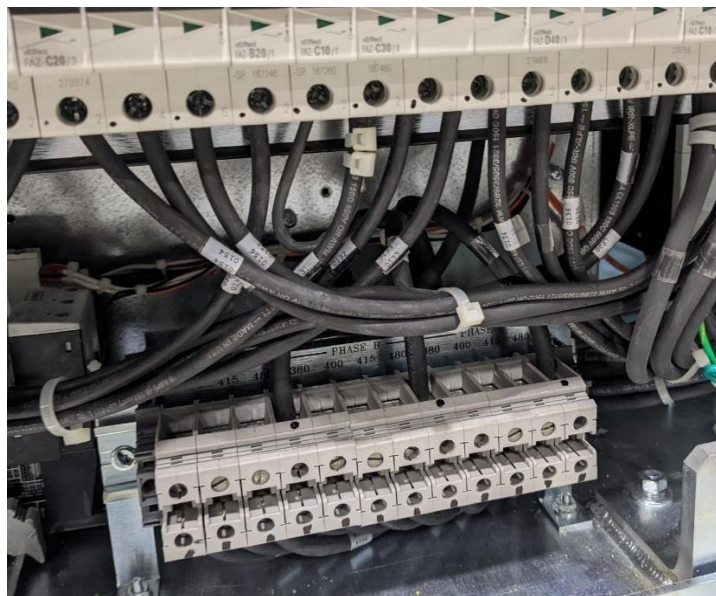
Unit is to be used solely for the purpose in which it was designed.

Unité doit être utilisé uniquement dans le but dans lequel il a été conçu.

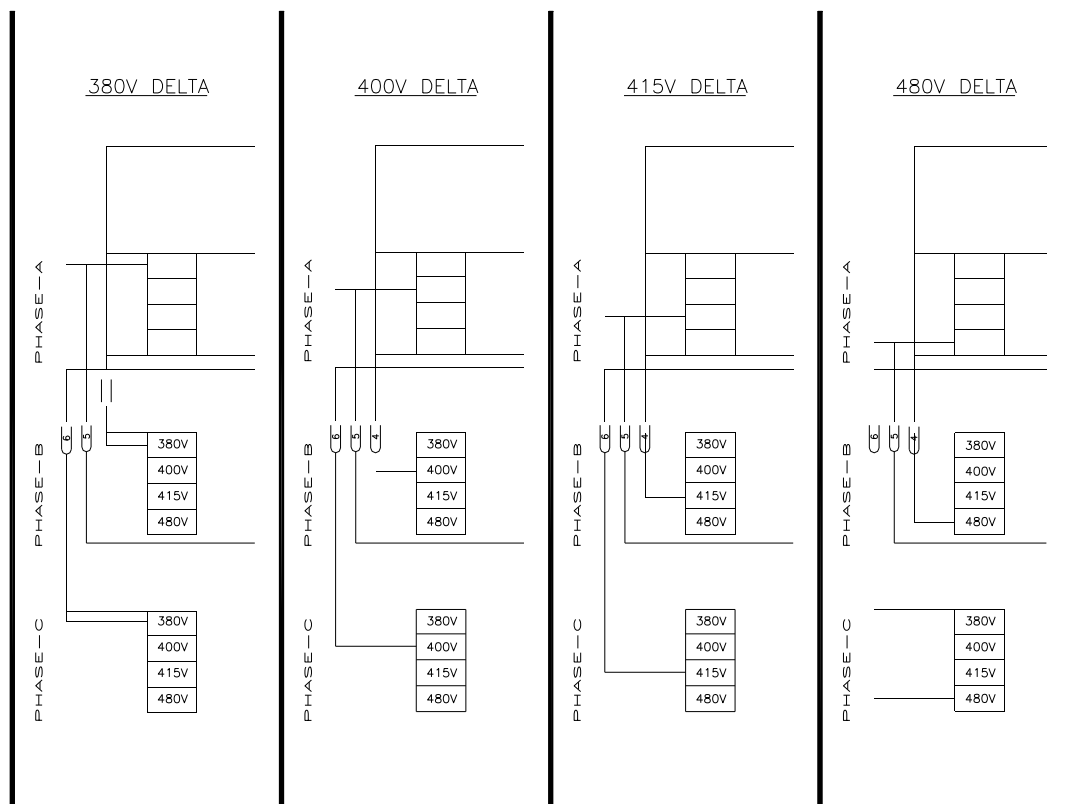
3.1.1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock out and tag out appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the small cover panels from the front panel of the PDU marked "INPUT VOLTAGE SELECT".
4. Illustration 3-1 shows the Input Voltage Selection terminal block (TB1). The illustration shows TB1 configured for 480V Delta. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK (TB1)
Illustration 3-1



INPUT VOLTAGE SELECTION
Illustration 3-2

3.1.2 Connection

Note the Cable Access Panel shown in Illustration 3-5 above the main circuit breaker. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.



For input cables, use wire sizes in accordance with local regulatory requirements for a 150A load

3.2 Operation

3.2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

OFF (as shown in Illustration 3-4),

ON (vertical), and

TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.



MAIN INPUT CIRCUIT BREAKER OPERATING HANDLE
Illustration 3-4

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3.2.2 Power Off

This pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3.2.3 EMO Reset

This pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3.2.4 TVSS Indicators

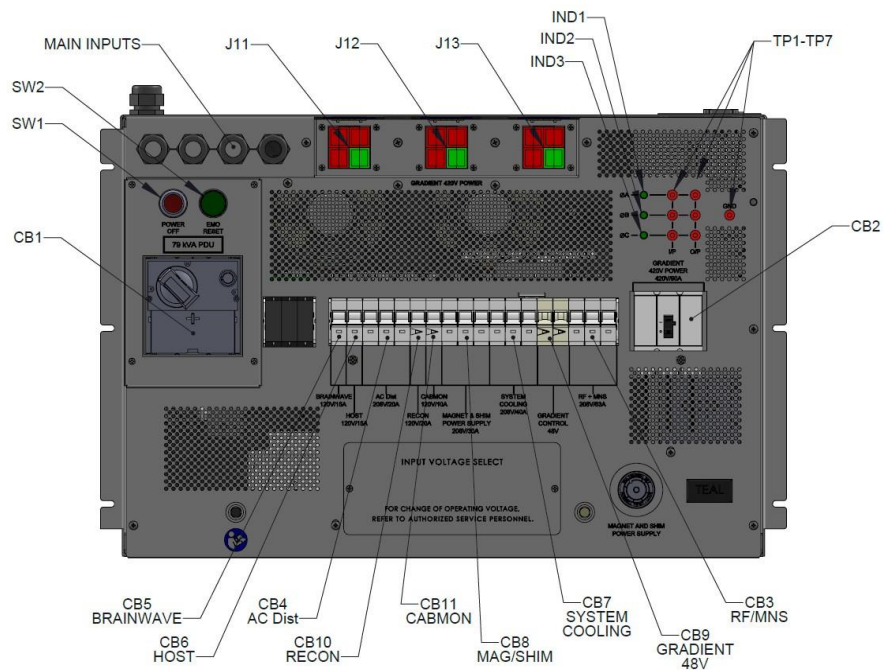
To the right of the Main Input Circuit Breaker are three TVSS (Transient Voltage Spike Suppressor). Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4.1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

3.2.5 Phase Indicators

There are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

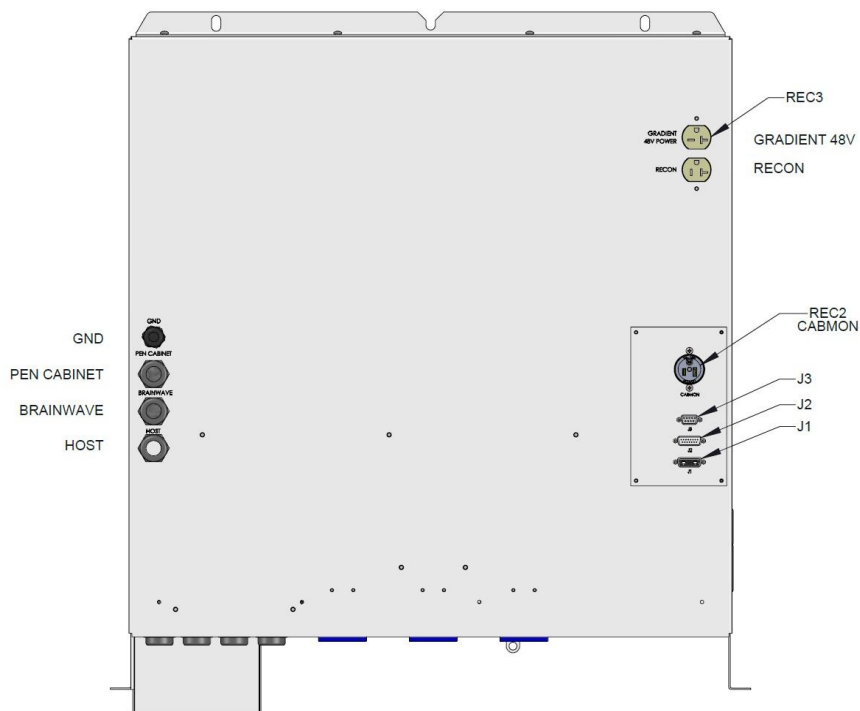
3.2.6 Input / Output Voltage Test Points

Near the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-5. The voltage at these points is reduced by a factor of 100:1 from the actual voltage at the terminal blocks.



FRONT PANEL COMPONENT LOCATIONS

Illustration 3-5



TOP PANEL COMPONENT LOCATIONS

Illustration 3-6

4- FIELD REPLACEABLE UNITS (FRUs)

No maintenance is required other than periodically (i.e. every 6 months) all electrical connections should be checked for tightness. Accumulations of dust or debris should be removed from front panel to keep ventilation working for convection cooling.

⚠ WARNING

All FRUs are to be replaced by authorized service personnel only.

Tous les FRUs doivent être remplacés par le personnel autorisé seulement.

To replace FRU components – first shutoff and lockout power, remove necessary panels, locate proper component (refer to Location drawing below) and replace with new component (refer to FRU table below). Reinstall all panels and restore power.

The exact type and rating fuse must be used as the replacement, or equipment damage may occur.

LISTING OF FRUs
4-1

Item	Ref Designator	Description	GE P/N	TEAL P/N
1	CB1	Main Input Circuit Breaker	5890114-10	4991587
2	CON1	PDU Gradient Contactor, 65A, 3-pole	5343114-4	4990950
3	CON2	PDU Estop Contactor, 95A, 3-pole	5890114-14	4991589
4	CON3	PDU System Contactor, 25A, 3-pole	5890114-15	4991591
5	CB2	Gradient 420V Power Circuit Breaker	5789505	4991592
6	R2	Relay for thermal switch	5890114-16	4991593
7	R1	Relay for main breaker trip	5890114-17	4991594
8	CB5-6, CB9-11	Brainwave, Host, Gradient 48V Control, Recon, and CabMon Circuit Breakers	5890114-18	4991595
9	CB3-4, CB7-8	RF-MNS, Pen Cabinet, System Cooling, and Magnet Power Supply Circuit Breakers	5890114-19	4991596
10	FAN1, FAN2	Fan With Tach Output Sanyo PN 109E1348G101	5140621-5	4990783
11	Q1-Q3	TVSS Module FRU for ACGD PDU	2301200-3	4990387
12	-	Gradient cable kit	5890114-20	4991597
13	IND1-3, SW1-2, TP1-7	Panel buttons, indicators, and test points	5789509	4991438

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
1.0	09/02/2022	Wang Jie	Extract Content from DOC2727916